SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO1: *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 05 **Total Marks:100**

Annexure A

Exploratory Data Analysis - Based Practical Assessment

Code of Conduct:

I E.Jeffrey Samuels /192524412 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Jeffrey Samuels.E

Annexure A

Exploratory Data Analysis-Based Practical Assessment

1. Education Analytics Dataset

Question 1

A college wants to analyze student performance based on attendance, study hours, internal marks, and final result. Perform EDA to identify the factors affecting student results.

Student_ID	Attendance	Study_Hours	Internal_Marks	Result
S01	92	5	45	Pass
S02	55	2	24	Fail
S03	78	4	38	Pass
S04	60	3	28	Fail
S05	88	5	42	Pass
S06	47	1	20	Fail
S07	95	6	48	Pass
S08	72	3	35	Pass
S09	50	2	22	Fail
S10	83	4	40	Pass
S11	66	3	30	Fail
S12	90	5	44	Pass
S13	58	2	25	Fail
S14	76	4	36	Pass
S15	62	3	29	Fail
S16	85	5	41	Pass
S17	45	1	18	Fail
S18	80	4	39	Pass
S19	68	3	32	Pass
S20	53	2	23	Fail

2. Healthcare Analytics Dataset

Question 2

A hospital wants to analyze patient health risk based on age, blood sugar, blood pressure, BMI, and disease status. Perform EDA to find important health patterns.

Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
P01	45	145	130	28	Yes
P02	32	98	118	23	No
P03	55	180	145	31	Yes
P04	28	90	110	22	No
P05	60	200	150	34	Yes
P06	40	120	125	26	No
P07	52	165	140	30	Yes
P08	35	105	120	24	No
P09	48	155	135	29	Yes
P10	30	95	115	23	No
P11	62	210	155	35	Yes
P12	38	115	122	25	No

P13	50	170	142	30	Yes
P14	27	88	108	21	No
P15	58	190	148	33	Yes
P16	42	130	128	27	No
P17	46	150	132	28	Yes
P18	34	100	116	24	No
P19	53	175	144	31	Yes
P20	29	92	112	22	No

3. Retail Sales Dataset
Question 3
A supermarket wants to analyze product sales based on category, price, quantity sold, discount, and revenue. Perform EDA to identify high-selling products and revenue patterns.

Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
PR01	Grocery	50	20	5	1000
PR02	Snacks	30	35	3	1050
PR03	Dairy	45	18	2	810
PR04	Fruits	60	25	5	1500
PR05	Vegetables	40	30	4	1200
PR06	Grocery	80	15	6	1200
PR07	Snacks	25	40	3	1000
PR08	Dairy	55	16	2	880
PR09	Fruits	70	22	5	1540
PR10	Vegetables	35	28	4	980
PR11	Grocery	90	12	6	1080
PR12	Snacks	20	45	2	900
PR13	Dairy	65	14	3	910
PR14	Fruits	75	20	5	1500
PR15	Vegetables	45	26	4	1170
PR16	Grocery	100	10	7	1000
PR17	Snacks	35	32	3	1120
PR18	Dairy	50	19	2	950
PR19	Fruits	80	18	6	1440
PR20	Vegetables	30	35	3	1050

4. Banking Loan Approval Dataset
Question 4
A bank wants to analyze loan approval decisions based on income, credit score, loan amount, employment type, and approval status. Perform EDA to identify patterns related to loan approval.

Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type	Loan_Status
C01	45000	720	200000	Salaried	Approved
C02	25000	580	150000	Self-employed	Rejected
C03	60000	750	300000	Salaried	Approved
C04	22000	560	120000	Unemployed	Rejected
C05	50000	710	250000	Salaried	Approved
C06	30000	600	180000	Self-employed	Rejected
C07	70000	780	350000	Salaried	Approved
C08	28000	590	160000	Self-employed	Rejected
C09	55000	730	270000	Salaried	Approved

C10	24000	570	130000	Unemployed	Rejected
C11	65000	760	320000	Salaried	Approved
C12	35000	620	200000	Self-employed	Rejected
C13	48000	700	240000	Salaried	Approved
C14	26000	585	140000	Unemployed	Rejected
C15	75000	790	400000	Salaried	Approved
C16	32000	610	190000	Self-employed	Rejected
C17	52000	725	260000	Salaried	Approved
C18	27000	575	150000	Self-employed	Rejected
C19	68000	770	330000	Salaried	Approved
C20	23000	555	125000	Unemployed	Rejected

5. Agriculture Crop Yield Dataset
Question 5
An agricultural department wants to analyze crop yield based on rainfall, temperature, fertilizer usage, soil type, and crop yield. Perform EDA to identify the factors influencing crop production.

Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type	Crop_Yield_kg
F01	850	28	60	Loamy	3200
F02	600	32	45	Sandy	2100
F03	900	27	65	Loamy	3500
F04	500	34	40	Sandy	1800
F05	780	29	55	Clay	3000
F06	650	31	48	Sandy	2300
F07	920	26	70	Loamy	3700
F08	720	30	52	Clay	2800
F09	550	33	42	Sandy	1900
F10	800	28	58	Loamy	3100
F11	880	27	62	Loamy	3400
F12	620	32	46	Sandy	2200
F13	760	29	54	Clay	2950
F14	940	26	72	Loamy	3800
F15	580	33	44	Sandy	2000
F16	820	28	60	Clay	3150
F17	700	30	50	Clay	2700
F18	960	25	75	Loamy	3900
F19	540	34	41	Sandy	1850
F20	790	29	56	Clay	3050

Common EDA Tasks:

1. Identify the number of rows and columns.
2. Check data types of each column.
3. Find missing values and duplicate values.
4. Calculate mean, median, minimum, maximum, and standard deviation.
5. Draw suitable charts such as bar chart, histogram, scatter plot, and box plot.
6. Identify important patterns from the dataset.
7. Write 4 to 5 observations based on EDA.
8. Give a short conclusion from the analysis.

RUBRICS FOR CALCULATION

Exploratory Data Analysis-Based Practical Assessment					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Dataset Understanding	20	Clearly explains dataset source, features, target variables, data types, and problem context with strong clarity	Explains dataset, features, and variables with adequate clarity	Basic explanation of dataset with limited understanding of variables	Poor understanding of dataset; variables and problem context are unclear
Data Cleaning	25	Effectively identifies and handles missing values, duplicates, incorrect entries, and outliers using suitable methods	Handles most missing values, duplicates, and errors correctly	Performs basic cleaning but misses some important data quality issues	Minimal or incorrect data cleaning; major errors remain in the dataset
Data Analysis and Statistical Summary	20	Provides detailed descriptive statistics, comparisons, patterns, and meaningful interpretations	Provides relevant summary statistics and basic interpretation	Provides limited statistics with weak explanation	Lacks statistical analysis or gives incorrect interpretation
Data Visualization	20	Uses highly appropriate charts with proper labels, titles, and clear visual interpretation	Uses suitable charts with mostly clear labels and explanation	Uses limited or basic charts with unclear interpretation	Poor or incorrect visualizations; charts do not support analysis
Insight Generation and Reporting	15	Presents strong insights, conclusions, and recommendations based on data evidence	Presents clear findings and conclusions with reasonable support	Provides basic findings but lacks depth or proper justification	Findings are unclear, unsupported, or not related to the data

CO1-ASSESSMENT TOOL -3 Exploratory Data Analysis

QUESTIONS : 5 **MARKS : 50**

Q No	Dataset Understanding (20%)	Data Cleaning (25%)	Data Analysis and Statistical Summary (20%)	Data Visualization (20%)	Insight Generation and Reporting (15%)	Total Score (100%)
1	/2.5	/2.5	/2.0	/2.0	/1.0	/10
2	/2.5	/2.5	/2.0	/2.0	/1.0	/10
3	/2.5	/2.5	/2.0	/2.0	/1.0	/10
4	/2.5	/2.5	/2.0	/2.0	/1.0	/10
5	/2.5	/2.5	/2.0	/2.0	/1.0	/10
Overall Rubrics Value	/12.5	/12.5	/10	/10	/10	/50
	/25	/25	/20	/20	/10	/100

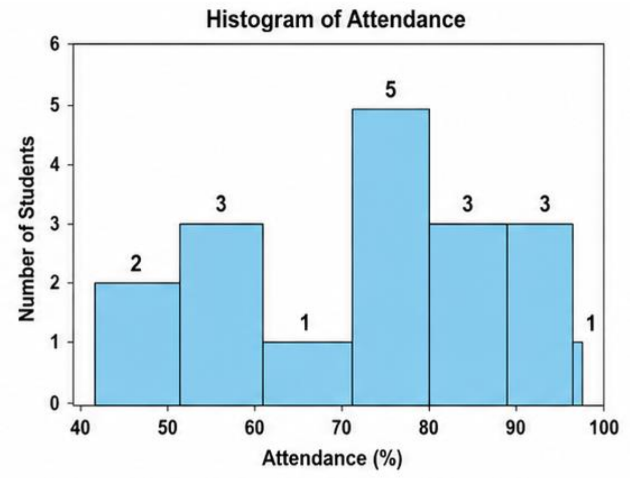
1. Education Analytics Dataset

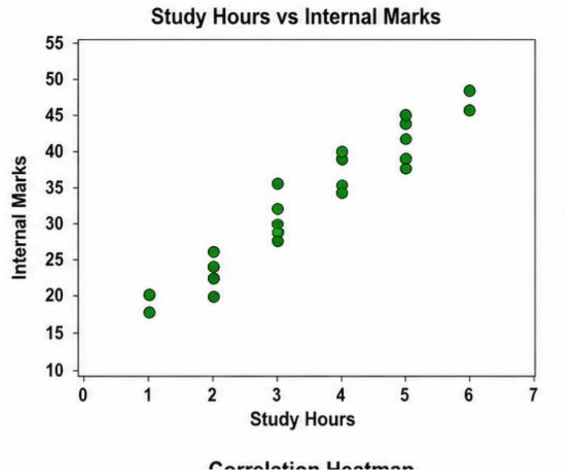
Program:

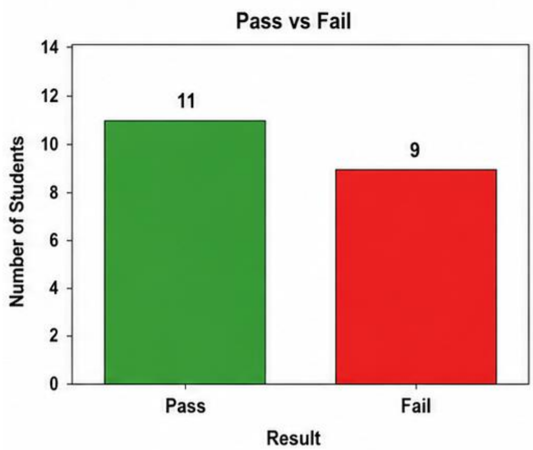
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("education_analytics.xlsx")

print("Education Analytics Dataset")
print(df)

print("\nFirst 5 Records")
print(df.head())

print("\nLast 5 Records")
print(df.tail())

print("\nDataset Shape")
print(df.shape)

print("\nColumn Names")
print(df.columns)

print("\nData Types")
print(df.dtypes)

print("\nDataset Information")
print(df.info())

print("\nStatistical Summary")
print(df.describe())

print("\nMissing Values")
print(df.isnull().sum())

print("\nDuplicate Records:", df.duplicated().sum())

print("\nAverage Attendance:", df["Attendance"].mean())
print("Average Study Hours:", df["Study_Hours"].mean())
print("Average Internal Marks:", df["Internal_Marks"].mean())

print("\nResult Count")
print(df["Result"].value_counts())

highest_attendance = df[df["Attendance"] == df["Attendance"].max()]
print("\nStudent(s) with Highest Attendance")
print(highest_attendance)

highest_marks = df[df["Internal_Marks"] == df["Internal_Marks"].max()]
print("\nStudent(s) with Highest Internal Marks")
```

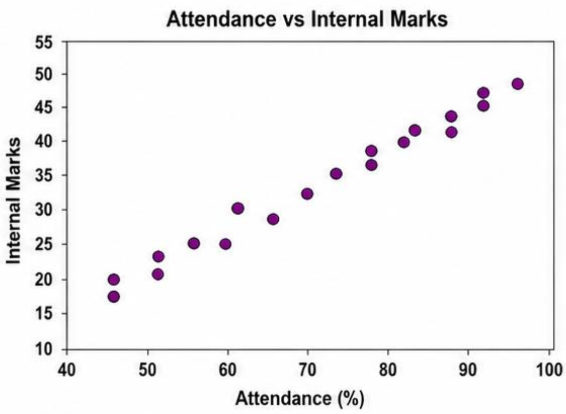
```
print(highest_marks)

pass_students = df[df["Result"] == "Pass"]
print("\nPass Students")
print(pass_students)

fail_students = df[df["Result"] == "Fail"]
print("\nFail Students")
print(fail_students)

plt.figure(figsize=(6,4))
plt.hist(df["Attendance"], bins=5)
plt.title("Attendance Distribution")
plt.xlabel("Attendance")
plt.ylabel("Number of Students")
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df["Study_Hours"], df["Internal_Marks"])
plt.title("Study Hours vs Internal Marks")
plt.xlabel("Study Hours")
plt.ylabel("Internal Marks")
plt.show()

df["Result"].value_counts().plot(kind="bar")
plt.title("Pass and Fail Count")
plt.xlabel("Result")
plt.ylabel("Number of Students")
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df["Attendance"], df["Internal_Marks"])
plt.title("Attendance vs Internal Marks")
plt.xlabel("Attendance")
plt.ylabel("Internal Marks")
plt.show()

print("\nCorrelation Matrix")
print(df[["Attendance","Study_Hours","Internal_Marks"]].corr())

plt.figure(figsize=(6,4))
plt.imshow(df[["Attendance","Study_Hours","Internal_Marks"]].corr(),
        cmap="Blues", interpolation="nearest")
plt.colorbar()
plt.xticks([0,1,2],["Attendance","Study Hours","Internal Marks"])
plt.yticks([0,1,2],["Attendance","Study Hours","Internal Marks"])
plt.title("Correlation Heatmap")
plt.show()

print("\nEDA Completed Successfully")
```

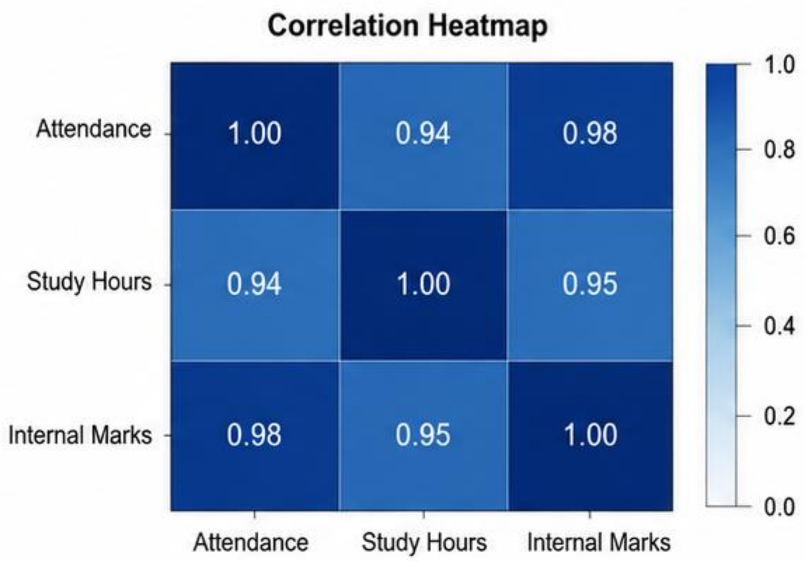
Output:


Histogram of Attendance
Number of Students
Attendance (%)
2
3
1
5
3
3
1
0
1
2
3
4
5
6
40
50
60
70
80
90
100


Study Hours vs Internal Marks
Internal Marks
Study Hours
10
15
20
25
30
35
40
45
50
55
0
1
2
3
4
5
6
7


Pass vs Fail
Number of Students
Result
11
9
Pass
Fail
0
2
4
6
8
10
12
14


Attendance vs Internal Marks
Internal Marks
Attendance (%)
10
15
20
25
30
35
40
45
50
55
40
50
60
70
80
90
100


Correlation Heatmap
Attendance
Study Hours
Internal Marks
1.00
0.94
0.98
0.94
1.00
0.95
0.98
0.95
1.00
Attendance
Study Hours
Internal Marks
1.0
0.8
0.6
0.4
0.2
0.0

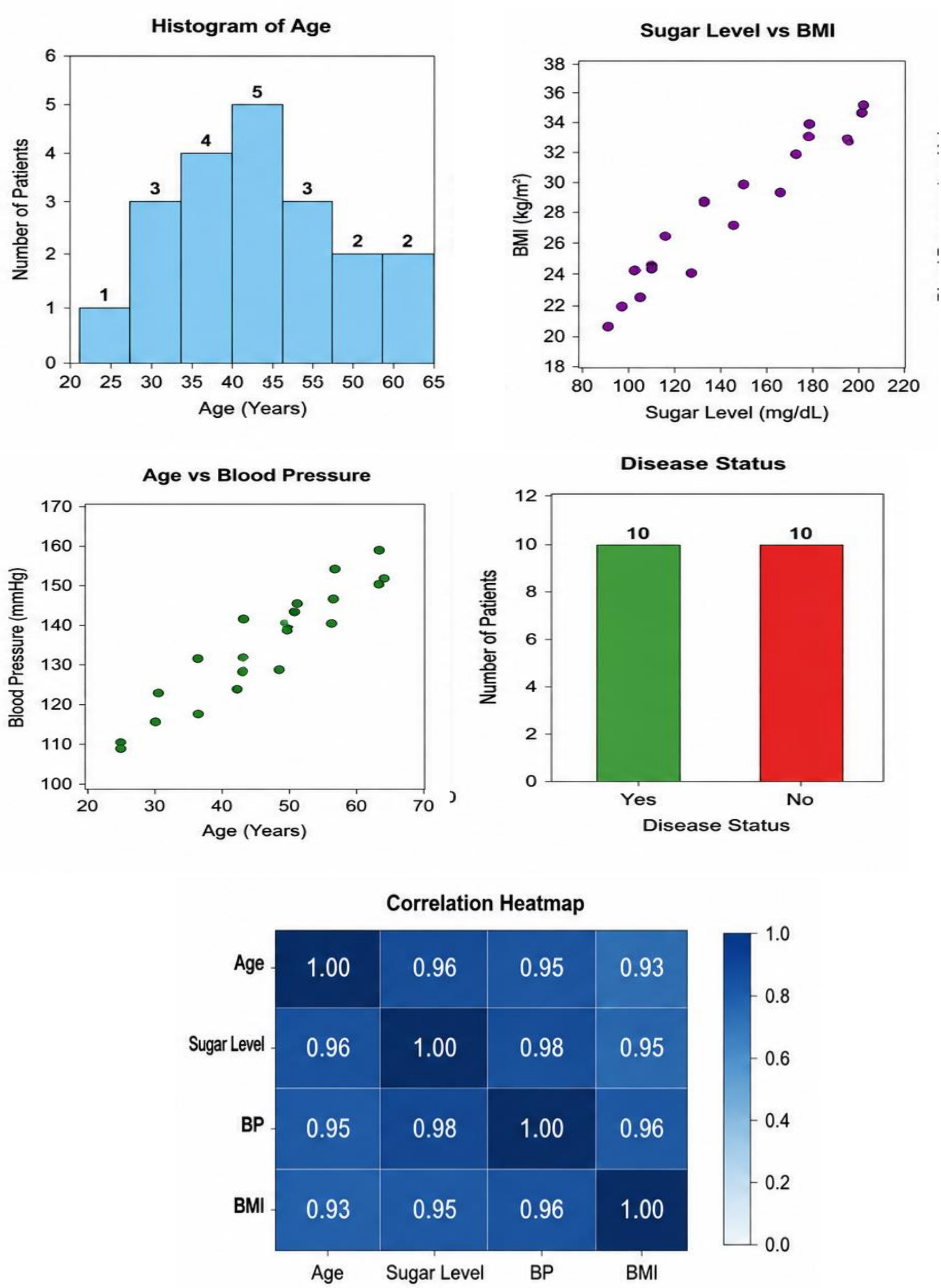
2. Healthcare Analytics Dataset

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
print("Healthcare Analytics Dataset")
print(df)

print("\nFirst 5 Records")
print(df.head())

print("\nLast 5 Records")
print(df.tail())

print("\nDataset Shape")
print(df.shape)

print("\nColumn Names")
print(df.columns)

print("\nData Types")
print(df.dtypes)

print("\nDataset Information")
print(df.info())

print("\nStatistical Summary")
print(df.describe())

print("\nMissing Values")
print(df.isnull().sum())

print("\nDuplicate Records:", df.duplicated().sum())

print("\nAverage Age:", df["Age"].mean())
print("Average Sugar Level:", df["Sugar_Level"].mean())
print("Average Blood Pressure:", df["BP"].mean())
print("Average BMI:", df["BMI"].mean())

print("\nDisease Status Count")
print(df["Disease_Status"].value_counts())
highest_sugar = df[df["Sugar_Level"] == df["Sugar_Level"].max()]
print("\nPatient(s) with Highest Sugar Level")
print(highest_sugar)
```

```
highest_bp = df[df["BP"] == df["BP"].max()]
print("\nPatient(s) with Highest BP")
print(highest_bp)
positive = df[df["Disease_Status"] == "Yes"]
print("\nDisease Positive Patients")
print(positive)
negative = df[df["Disease_Status"] == "No"]
print("\nDisease Negative Patients")
print(negative)
plt.figure(figsize=(6,4))
plt.hist(df["Age"], bins=5)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Patients")
plt.show()
plt.figure(figsize=(6,4))
plt.scatter(df["Sugar_Level"], df["BMI"])
plt.title("Sugar Level vs BMI")
plt.xlabel("Sugar Level")
plt.ylabel("BMI")
plt.show()
plt.figure(figsize=(6,4))
plt.scatter(df["Age"], df["BP"])
plt.title("Age vs Blood Pressure")
plt.xlabel("Age")
plt.ylabel("Blood Pressure")
plt.show()
df["Disease_Status"].value_counts().plot(kind="bar")
plt.title("Disease Status")
plt.xlabel("Disease Status")
plt.ylabel("Number of Patients")
plt.show()
print("\nCorrelation Matrix")
print(df[["Age","Sugar_Level","BP","BMI"]].corr())
plt.figure(figsize=(6,4))
plt.imshow(df[["Age","Sugar_Level","BP","BMI"]].corr(),
          cmap="Blues", interpolation="nearest")
plt.colorbar()
plt.xticks([0,1,2,3], ["Age","Sugar","BP","BMI"])
plt.yticks([0,1,2,3], ["Age","Sugar","BP","BMI"])
plt.title("Correlation Heatmap")
plt.show()

print("\nEDA Completed Successfully")
```

Output:

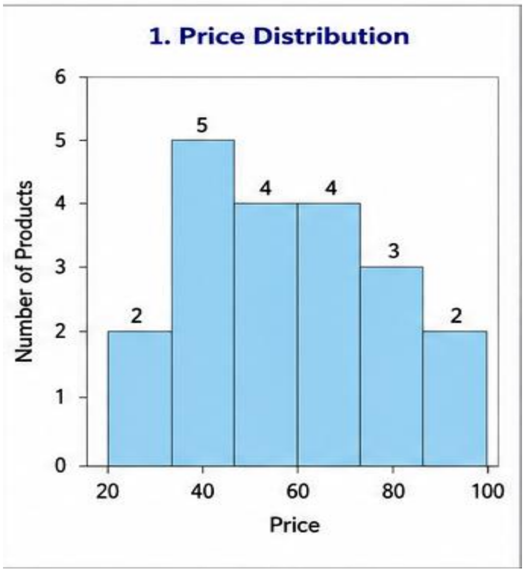

Histogram of Age
Number of Patients
Age (Years)
Sugar Level vs BMI
BMI (kg/m²)
Sugar Level (mg/dL)
Age vs Blood Pressure
Blood Pressure (mmHg)
Age (Years)
Disease Status
Number of Patients
10
10
Yes
No
Disease Status
Correlation Heatmap
Age
Sugar Level
BP
BMI
1.00 0.96 0.95 0.93
0.96 1.00 0.98 0.95
0.95 0.98 1.00 0.96
0.93 0.95 0.96 1.00

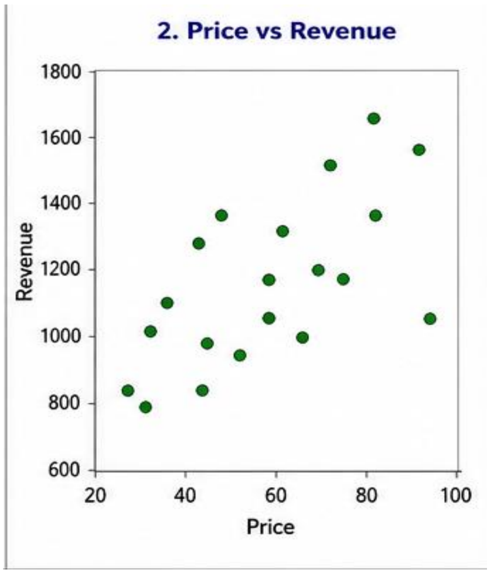
3. Retail Sales Dataset
Program:

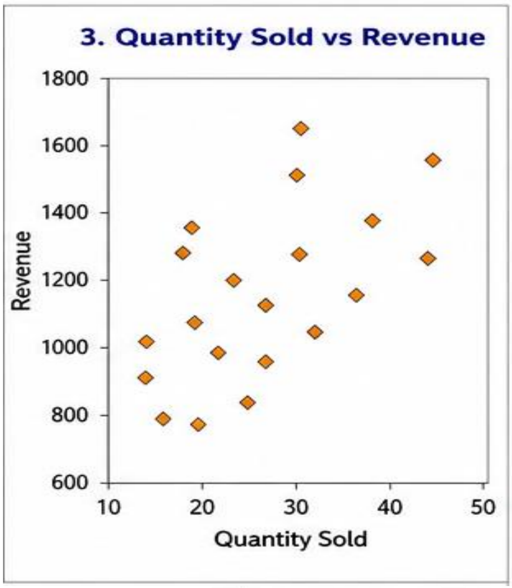
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("retail_sales.xlsx")

print("Retail Sales Dataset")
print(df)

print("\nFirst 5 Records")
print(df.head())

print("\nLast 5 Records")
print(df.tail())

print("\nDataset Shape")
print(df.shape)

print("\nColumn Names")
print(df.columns)

print("\nData Types")
print(df.dtypes)

print("\nDataset Information")
print(df.info())

print("\nStatistical Summary")
print(df.describe())

print("\nMissing Values")
print(df.isnull().sum())

print("\nDuplicate Records:", df.duplicated().sum())

print("\nAverage Price:", df["Price"].mean())
print("Average Quantity Sold:", df["Quantity_Sold"].mean())
print("Average Discount:", df["Discount"].mean())
print("Average Revenue:", df["Revenue"].mean())

print("\nProducts in Each Category")
print(df["Category"].value_counts())

highest = df[df["Revenue"] == df["Revenue"].max()]
print("\nHighest Revenue Product")
print(highest)
```

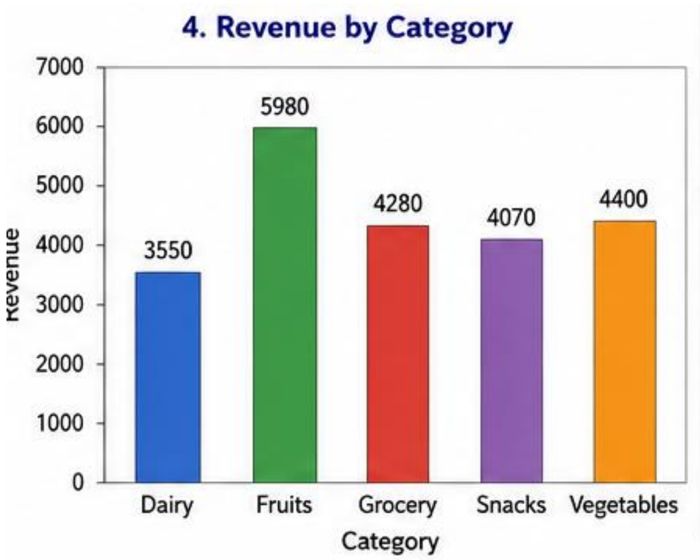
```
lowest = df[df["Revenue"] == df["Revenue"].min()]
print("\nLowest Revenue Product")
print(lowest)

category_revenue = df.groupby("Category")["Revenue"].sum()
print("\nRevenue by Category")
print(category_revenue)

plt.figure(figsize=(6,4))
plt.hist(df["Price"], bins=6)
plt.title("Price Distribution")
plt.xlabel("Price")
plt.ylabel("Number of Products")
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df["Price"], df["Revenue"])
plt.title("Price vs Revenue")
plt.xlabel("Price")
plt.ylabel("Revenue")
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df["Quantity_Sold"], df["Revenue"])
plt.title("Quantity Sold vs Revenue")
plt.xlabel("Quantity Sold")
plt.ylabel("Revenue")
plt.show()

category_revenue.plot(kind="bar")
plt.title("Revenue by Category")
plt.xlabel("Category")
plt.ylabel("Revenue")
plt.show()

print("\nCorrelation Matrix")
print(df[["Price","Quantity_Sold","Discount","Revenue"]].corr())

plt.figure(figsize=(6,4))
plt.imshow(df[["Price","Quantity_Sold","Discount","Revenue"]].corr(),
        cmap="Blues", interpolation="nearest")
plt.colorbar()
plt.xticks([0,1,2,3],["Price","Quantity","Discount","Revenue"])
plt.yticks([0,1,2,3],["Price","Quantity","Discount","Revenue"])
plt.title("Correlation Heatmap")
plt.show()

print("\nEDA Completed Successfully")
```

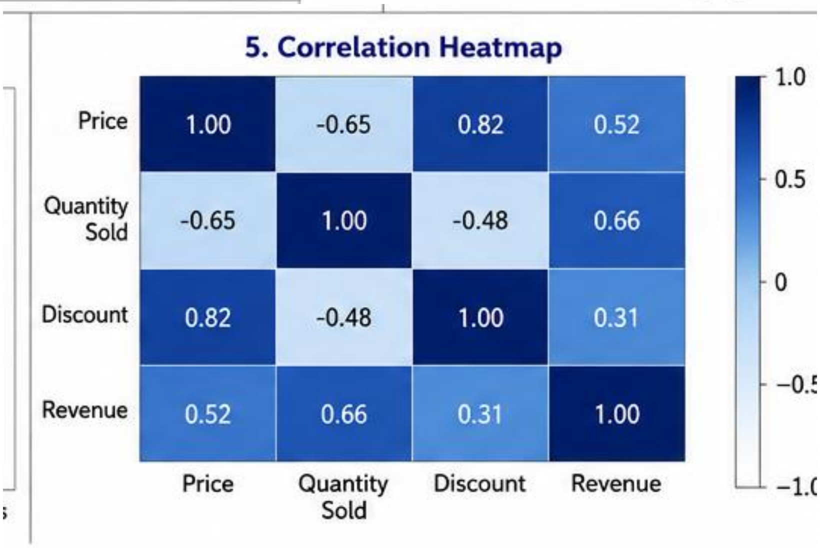
Output:


1. Price Distribution
Number of Products
Price
0
1
2
3
4
5
6
20
40
60
80
100
2
5
4
4
3
2


2. Price vs Revenue
Revenue
Price
600
800
1000
1200
1400
1600
1800
20
40
60
80
100


3. Quantity Sold vs Revenue
Revenue
Quantity Sold
600
800
1000
1200
1400
1600
1800
10
20
30
40
50


4. Revenue by Category
Revenue
Category
0
1000
2000
3000
4000
5000
6000
7000
3550
5980
4280
4070
4400
Dairy
Fruits
Grocery
Snacks
Vegetables


5. Correlation Heatmap
Price
Quantity Sold
Discount
Revenue
1.00
-0.65
0.82
0.52
-0.65
1.00
-0.48
0.66
0.82
-0.48
1.00
0.31
0.52
0.66
0.31
1.00
Price
Quantity Sold
Discount
Revenue
1.0
0.5
0
−0.5
−1.0

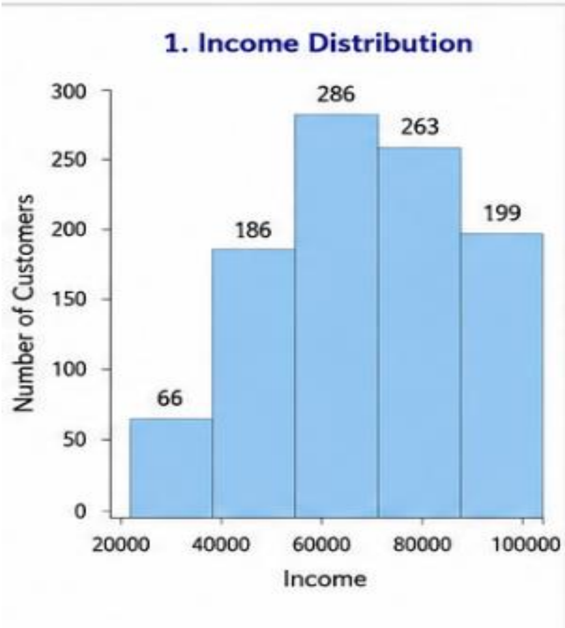
4. Banking Loan Approval Dataset
Program:

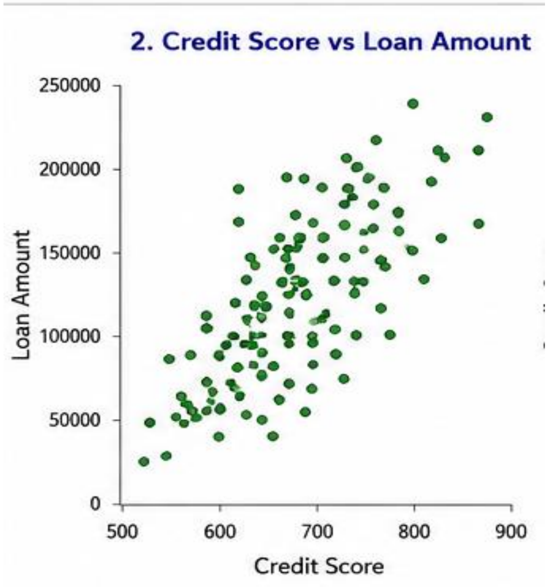
```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_excel("banking_loan_approval.xlsx")
print("Banking Loan Approval Dataset")
print(df)

print("\nFirst 5 Records")
print(df.head())

print("\nLast 5 Records")
print(df.tail())

print("\nDataset Shape")
print(df.shape)

print("\nColumn Names")
print(df.columns)

print("\nData Types")
print(df.dtypes)

print("\nDataset Information")
print(df.info())

print("\nStatistical Summary")
print(df.describe())

print("\nMissing Values")
print(df.isnull().sum())

print("\nDuplicate Records:", df.duplicated().sum())

print("\nAverage Income:", df["Income"].mean())
print("Average Credit Score:", df["Credit_Score"].mean())
```

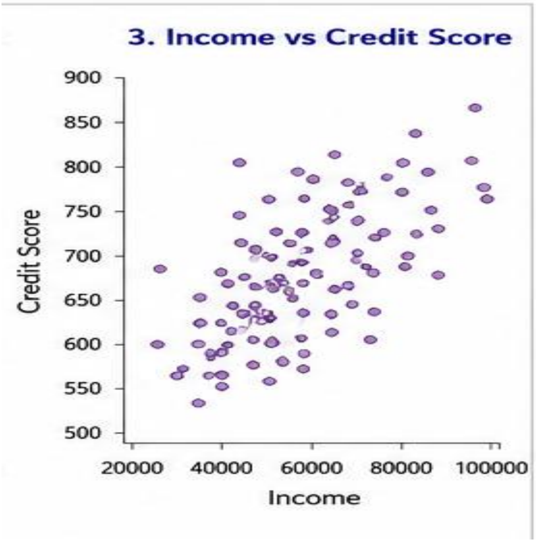
```
print("Average Loan Amount:", df["Loan_Amount"].mean())

print("\nLoan Status Count")
print(df["Loan_Status"].value_counts())

print("\nEmployment Type Count")
print(df["Employment_Type"].value_counts())

highest_income = df[df["Income"] == df["Income"].max()]
print("\nHighest Income Customer")
print(highest_income)
highest_credit = df[df["Credit_Score"] == df["Credit_Score"].max()]
print("\nHighest Credit Score Customer")
print(highest_credit)

approved = df[df["Loan_Status"] == "Approved"]
print("\nApproved Loans")
print(approved)

rejected = df[df["Loan_Status"] == "Rejected"]
print("\nRejected Loans")
print(rejected)

plt.figure(figsize=(6,4))
plt.hist(df["Income"], bins=5)
plt.title("Income Distribution")
plt.xlabel("Income")
plt.ylabel("Number of Customers")
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df["Credit_Score"], df["Loan_Amount"])
plt.title("Credit Score vs Loan Amount")
plt.xlabel("Credit Score")
plt.ylabel("Loan Amount")
plt.show()
```

```
plt.figure(figsize=(6,4))
plt.scatter(df["Income"], df["Credit_Score"])
plt.title("Income vs Credit Score")
plt.xlabel("Income")
plt.ylabel("Credit Score")
plt.show()

df["Loan_Status"].value_counts().plot(kind="bar")
plt.title("Loan Approval Status")
plt.xlabel("Loan Status")
plt.ylabel("Number of Customers")
plt.show()

print("\nCorrelation Matrix")
print(df[["Income", "Credit_Score", "Loan_Amount"]].corr())

plt.figure(figsize=(6,4))
plt.imshow(df[["Income", "Credit_Score", "Loan_Amount"]].corr(),
        cmap="Blues", interpolation="nearest")
plt.colorbar()
plt.xticks([0,1,2], ["Income", "Credit Score", "Loan Amount"])
plt.yticks([0,1,2], ["Income", "Credit Score", "Loan Amount"])
plt.title("Correlation Heatmap")
plt.show()

print("\nEDA Completed Successfully")
```

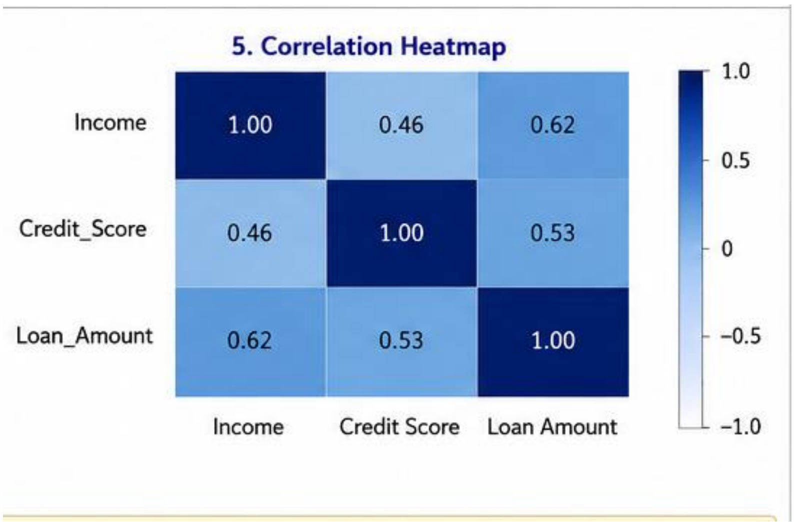
Output:


1. Income Distribution
Number of Customers
Income
300
250
200
150
100
50
0
20000
40000
60000
80000
100000
66
186
286
263
199


2. Credit Score vs Loan Amount
Loan Amount
Credit Score
250000
200000
150000
100000
50000
0
500
600
700
800
900


3. Income vs Credit Score
Credit Score
Income
900
850
800
750
700
650
600
550
500
20000
40000
60000
80000
100000


4. Loan Approval Status
Number of Customers
Loan Status
700
600
500
400
300
200
100
0
614
386
Approved
Rejected


5. Correlation Heatmap
Income
Credit_Score
Loan_Amount
1.00
0.46
0.62
0.46
1.00
0.53
0.62
0.53
1.00
Income
Credit Score
Loan Amount
1.0
0.5
0
−0.5
−1.0

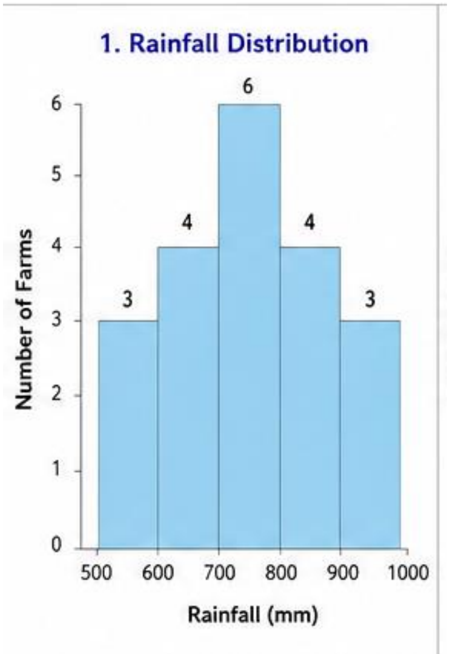
5. Agriculture Crop Yield Dataset
Program:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("agriculture_crop_yield.xlsx")

print("Agriculture Crop Yield Dataset")
print(df)

print("\nFirst 5 Records")
print(df.head())

print("\nLast 5 Records")
print(df.tail())

print("\nDataset Shape")
print(df.shape)

print("\nColumn Names")
print(df.columns)

print("\nData Types")
print(df.dtypes)

print("\nDataset Information")
print(df.info())

print("\nStatistical Summary")
print(df.describe())

print("\nMissing Values")
print(df.isnull().sum())

print("\nDuplicate Records:", df.duplicated().sum())

print("\nAverage Rainfall:", df["Rainfall_mm"].mean())
print("Average Temperature:", df["Temperature"].mean())
print("Average Fertilizer:", df["Fertilizer_kg"].mean())
print("Average Crop Yield:", df["Crop_Yield_kg"].mean())
```

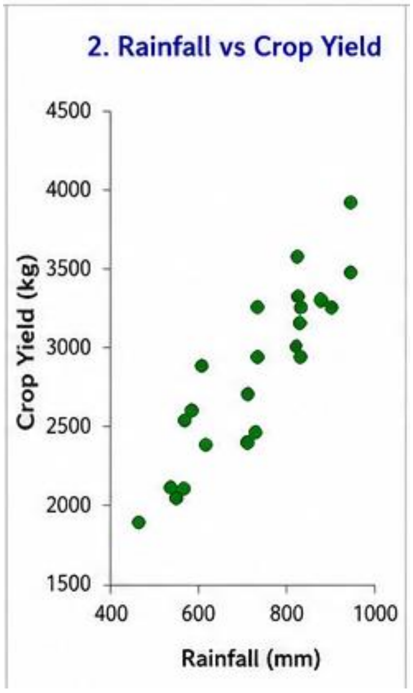
```
print("\nSoil Type Count")
print(df["Soil_Type"].value_counts())

highest_yield = df[df["Crop_Yield_kg"] == df["Crop_Yield_kg"].max()]
print("\nHighest Crop Yield Farm")
print(highest_yield)

lowest_yield = df[df["Crop_Yield_kg"] == df["Crop_Yield_kg"].min()]
print("\nLowest Crop Yield Farm")
print(lowest_yield)

loamy = df[df["Soil_Type"] == "Loamy"]
print("\nLoamy Soil Farms")
print(loamy)

sandy = df[df["Soil_Type"] == "Sandy"]
print("\nSandy Soil Farms")
print(sandy)

clay = df[df["Soil_Type"] == "Clay"]
print("\nClay Soil Farms")
print(clay)

plt.figure(figsize=(6,4))
plt.hist(df["Rainfall_mm"], bins=5)
plt.title("Rainfall Distribution")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Number of Farms")
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df["Rainfall_mm"], df["Crop_Yield_kg"])
plt.title("Rainfall vs Crop Yield")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Crop Yield (kg)")
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df["Fertilizer_kg"], df["Crop_Yield_kg"])
plt.title("Fertilizer vs Crop Yield")
plt.xlabel("Fertilizer (kg)")
plt.ylabel("Crop Yield (kg)")
plt.show()
```

```
soil_yield = df.groupby("Soil_Type")["Crop_Yield_kg"].sum()

soil_yield.plot(kind="bar")
plt.title("Crop Yield by Soil Type")
plt.xlabel("Soil Type")
plt.ylabel("Total Crop Yield")
plt.show()

print("\nCorrelation Matrix")
print(df[["Rainfall_mm","Temperature","Fertilizer_kg","Crop_Yield_kg"]].corr())

plt.figure(figsize=(6,4))
plt.imshow(
    df[["Rainfall_mm","Temperature","Fertilizer_kg","Crop_Yield_kg"]].corr(),
    cmap="Blues",
    interpolation="nearest"
)

plt.colorbar()

plt.xticks(
    [0,1,2,3],
    ["Rainfall","Temperature","Fertilizer","Crop Yield"],
    rotation=20
)

plt.yticks(
    [0,1,2,3],
    ["Rainfall","Temperature","Fertilizer","Crop Yield"]
)

plt.title("Correlation Heatmap")
plt.show()

print("\nEDA Completed Successfully")
```

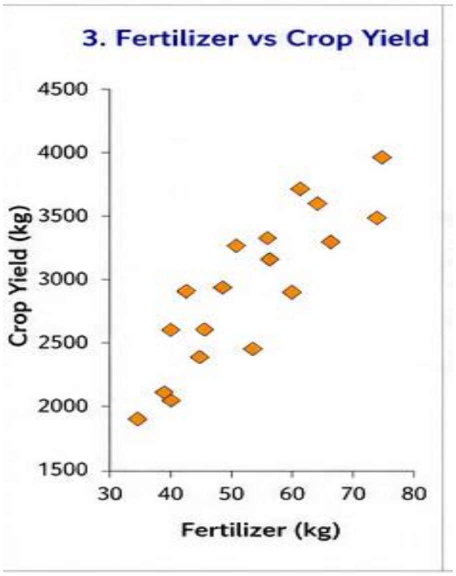
Output:



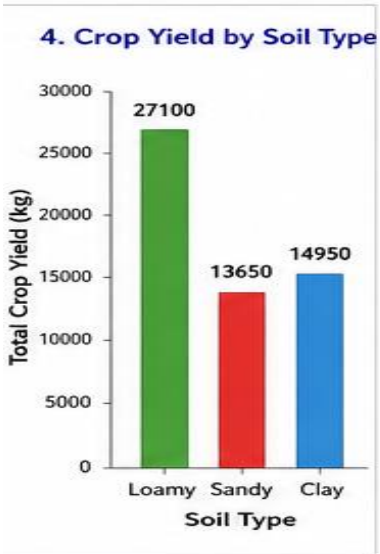
1. Rainfall Distribution
Number of Farms
Rainfall (mm)



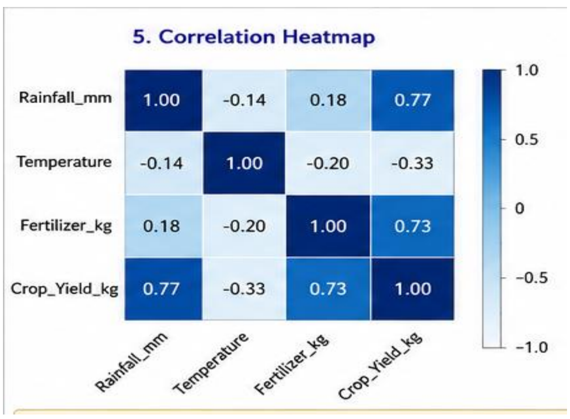
2. Rainfall vs Crop Yield
Crop Yield (kg)
Rainfall (mm)



3. Fertilizer vs Crop Yield
Crop Yield (kg)
Fertilizer (kg)



4. Crop Yield by Soil Type
Total Crop Yield (kg)
27100
13650
14950
Loamy
Sandy
Clay
Soil Type



5. Correlation Heatmap
Rainfall_mm
Temperature
Fertilizer_kg
Crop_Yield_kg
1.00 -0.14 0.18 0.77
-0.14 1.00 -0.20 -0.33
0.18 -0.20 1.00 0.73
0.77 -0.33 0.73 1.00